

1. Given

$$f(x) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)},$$

where $x, \mu \in \mathbb{R}^k$, Σ is a k -by- k positive definite matrix and $|\Sigma|$ is its determinant.

Show that $\int_{\mathbb{R}^k} f(x) dx = 1$.

(pf) Let $y = x - \mu$, $dy = dx$

$$\int_{\mathbb{R}^k} f(x) dx = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \int_{\mathbb{R}^k} \exp\left(-\frac{1}{2} y^T \Sigma y\right) dy$$

$$\text{so we prove } \int_{\mathbb{R}^k} \exp\left(-\frac{1}{2} y^T \Sigma y\right) dy = \sqrt{(2\pi)^k |\Sigma|} \Rightarrow \int_{\mathbb{R}^k} f(x) dx = 1$$

Σ is positive definite matrix, by Cholesky decomposition

$$\Sigma = LL^T, |\Sigma| = |L|^2, \text{ and let } z = L^{-1}y, dy = |L| dz$$

$$y^T \Sigma^{-1} y = y^T (L^T)^{-1} L^{-1} y = \underset{(L^T)^{-1} = (L^{-1})^T}{z^T z}$$

$$\int_{\mathbb{R}^k} f(x) dx = |L| \int_{\mathbb{R}^k} \exp\left(-\frac{1}{2} \|z\|^2\right) dz = |L| \int_{\mathbb{R}^k} \exp\left(-\frac{1}{2} \sum_{i=1}^k z_i^2\right) dz$$

$$= \prod_{i=1}^k \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2} z_i^2\right) dz_i = \left(\sqrt{2\pi}\right)^k$$
$$\int_{-\infty}^{\infty} e^{-u^2/2} = \sqrt{2\pi}$$

$$\text{so } \int_{\mathbb{R}^k} \exp\left(-\frac{1}{2} y^T \Sigma^{-1} y\right) dy = |L| (2\pi)^{k/2} = \sqrt{|\Sigma|} (2\pi)^{k/2} = \sqrt{(2\pi)^k |\Sigma|}$$

$$\int_{\mathbb{R}} f(x) dx = \frac{\sqrt{(2\pi)^k |\Sigma|}}{\sqrt{(2\pi)^k |\Sigma|}} = 1$$

2. Let A, B be n -by- n matrices and x be a n -by-1 vector.

(a) Show that $\frac{\partial}{\partial A} \text{trace}(AB) = B^T$.

(b) Show that $x^T A x = \text{trace}(x x^T A)$.

(b) Derive the maximum likelihood estimators for a multivariate Gaussian.

$$(a) \text{trace}(AB) = \sum_i (AB)_{ii} = \sum_i \left(\sum_j A_{ij} B_{ji} \right)$$

$$\frac{\partial}{\partial A_{kI}} \text{trace}(AB) = \frac{\partial}{\partial A_{kI}} \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ji} = B_{Ik}$$

$$B_{Ik} = (B^T)_{kI}, \quad \frac{\partial}{\partial A} \text{trace}(AB) = B^T$$

$$(b) x^T A x = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j$$

$$(x x^T A)_{mm} = \sum_p (x x^T)_{mp} A_{pm} = \sum_p x_m x_p A_{pm}$$

$$\Rightarrow \text{trace}(x x^T A) = \sum_m \sum_p x_m x_p A_{pm} = \sum_{i=1}^n \sum_{j=1}^n x_i A_{ij} x_j = x^T A x$$

(c) unknown parameter: μ, Σ

$$L(\mu, \Sigma) = \prod_{i=1}^m f(x_i) = (2\pi)^{-\frac{km}{2}} |\Sigma|^{-\frac{m}{2}} \exp\left(-\frac{1}{2} \sum_{i=1}^m (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)\right)$$

$$\ln(L) = -\frac{km}{2} \log(2\pi) - \frac{m}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^m (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)$$

$$\frac{\partial \ln(L)}{\partial \mu} = -\frac{1}{2} \left(\frac{\partial}{\partial \mu} \left(\sum_{i=1}^m \|x_i - \mu\|_{\Sigma^{-1}}^2 \right) \right) = \Sigma^{-1} \cdot \sum_i (x_i - \mu) = 0 \Rightarrow \mu = \bar{x}$$

$$\text{Let } S = \frac{1}{m} \sum_i (x_i - \bar{x})(x_i - \bar{x})^T$$

$$l(\Sigma) = C - \frac{m}{2} \log |\Sigma| - \frac{1}{2} \text{trace}(\Sigma^{-1} m S)$$

$$\frac{\partial l(\Sigma)}{\partial \Sigma} = -\frac{m}{2} \Sigma^{-1} + \frac{m}{2} \Sigma^{-1} S \Sigma^{-1} = 0 \quad \left(\frac{\partial}{\partial \Sigma} \text{trace}(\Sigma^{-1} S) = -\Sigma^{-1} S \Sigma^{-1} \right)$$

$$\Rightarrow \Sigma^{-1} = \Sigma^{-1} S \Sigma^{-1} \Rightarrow \Sigma = S = \frac{1}{m} \sum_{i=1}^m (x_i - \bar{x})(x_i - \bar{x})^T$$

(3) unknown problem

一開始修機器學習，都會想做一個程式

可以幫我打遊戲(每日任務等等)，但近期

學習的內容感覺做不出來。感覺不是一個

線性函數可以做到的，但不確定 Assignment 4

的 random forest 有機會(?) 動態遊戲可能

沒辦法，不過象棋、西洋棋的可能有機會
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